SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Computer Networks: OSI/TCP-IP Protocols, Routing & Congestion Control

Department: CSE	**Course / Subject:** Computer Networks (CS502PC)
Semester: Semester 5	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Layered protocol stacks, Data link framing & CRC, Sliding Window protocols, Routing algorithms (OSPF, BGP), TCP Reno/Tahoe, and Application layer protocols (DNS, HTTP/2).

Unit 1: Network Physical & Data Link Layer

OSI 7-layer reference model, TCP/IP architecture, Framing, Flow control (Stop-and-Wait, Go-Back-N, Selective Repeat), Error detection (CRC, Checksum).

Unit 2: Medium Access Control (MAC)

ALOHA (Pure and Slotted), CSMA/CD (Ethernet IEEE 802.3), CSMA/CA (Wi-Fi IEEE 802.11), and Switching methods (Virtual circuit vs Datagram).

Unit 3: Network Layer & Routing

IPv4 addressing, Subnetting, CIDR, IPv6 header format, Distance Vector Routing (Bellman-Ford), Link State Routing (Dijkstra/OSPF), and BGP routing.

Unit 4: Transport Layer

User Datagram Protocol (UDP), Transmission Control Protocol (TCP), 3-way handshake, Flow control, TCP Congestion Control algorithms (AIMD, Slow Start).

Unit 5: Application Layer & Security

Domain Name System (DNS), HTTP/1.1 vs HTTP/2, SMTP, POP3, IMAP, SSL/TLS handshake overview, and symmetric/asymmetric cryptographic principles.